Hayder Alraies

Backend & Systems Software Engineer

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Summary

Backend & Systems Software Engineer specializing in Modern C++, Python, and Linux. Experienced in building high-performance backend services, concurrent systems, and production-ready infrastructure with a strong focus on performance, reliability, and clean software architecture.

Experience

Independent Software Engineer — Dec 2025 – present

- Independently designed, built, and shipped five backend and systems projects end-to-end, spanning Python/FastAPI services, Modern C++ systems programming, and real-time infrastructure — covering architecture, implementation, testing, and deployment.
- Applied production-oriented engineering practices including asynchronous programming, caching, containerization, and performance benchmarking to validate reliability and throughput under load.

Projects

BoltKV – Code Source — July 2026 – present

High-performance in-memory key-value database

- Designed and built a high-performance, thread-safe in-memory key-value database in Modern C++ featuring Linux `epoll` networking, concurrent request handling, append-only persistence (AOF), and crash recovery.
- Engineered a modular storage architecture with unit testing, CMake-based build system, and performance-focused design, demonstrating expertise in systems programming, memory management, concurrency, and Linux networking.
- Benchmarked at 23,390+ QPS with 0.0428ms average latency per request, sustaining 250,000 operations across 50 concurrent workers in 10.688 seconds.

Real-Time-API-Monitoring-Platform – Code Source — June 2026 – present

Production-inspired observability platform built with FastAPI, Redis, PostgreSQL, and WebSockets for monitoring API traffic, latency, and live system metrics.

- FastAPI backend with WebSocket support, Redis-based real-time metric collection.
- PostgreSQL persistence via SQLAlchemy async ORM, Docker Compose ready for local development and deployment.
- Processed over 205,000 requests in real time while maintaining an average latency of 172ms, tracked live on a Tailwind-powered dashboard UI.

QazVelo Engine – Live Demo — May 2026 – June 2026

Real-time financial analytics and mock trading platform

- Architected a high-throughput backend system utilizing FastAPI to process live cryptocurrency market data streams directly from Binance WebSockets.
- Designed a fully asynchronous database layer using PostgreSQL and Async SQLAlchemy, ensuring non-blocking, concurrent mock trading operations.
- Implemented distributed caching and strict rate-limiting mechanisms using Redis to maintain system stability and prevent bottlenecks under heavy data loads.
- Containerized the infrastructure using Docker and deployed on Railway, resolving complex CORS policies and API/WS proxy routing.

PulseNet – Live Demo — Apr 2026 – May 2026

Scalable backend application API

- Developed a robust backend application using FastAPI, focusing on clean API design, rapid request handling, and scalable architecture.
- Managed the full software development lifecycle, from initial logic design to configuring deployment environments.

GraphFlow-Lite – Code Source — Dec 2025 – Jan 2026

High-performance algorithmic core system

- Engineered a computational core system in C++ focused on optimal memory management and raw execution speed.
- Designed and implemented advanced data structures and algorithms to efficiently process and route complex data graphs, minimizing time complexity.
- Focused on writing clean, low-level, and highly optimized algorithmic code to ensure maximum hardware utilization.

Skills

Languages: Python, Modern C++ (C++17), SQL
Backend & Databases: FastAPI, PostgreSQL, Redis, REST APIs, WebSockets, SQLAlchemy
Systems Programming: Linux (Debian), TCP/IP Networking, epoll, Concurrency, Memory Management, Performance Optimization
DevOps & Infrastructure: Docker, Git, GitHub Actions, Railway, Vercel